4-Lepton Off-shell Higgs boson Production Measurements

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The off-shell Higgs boson

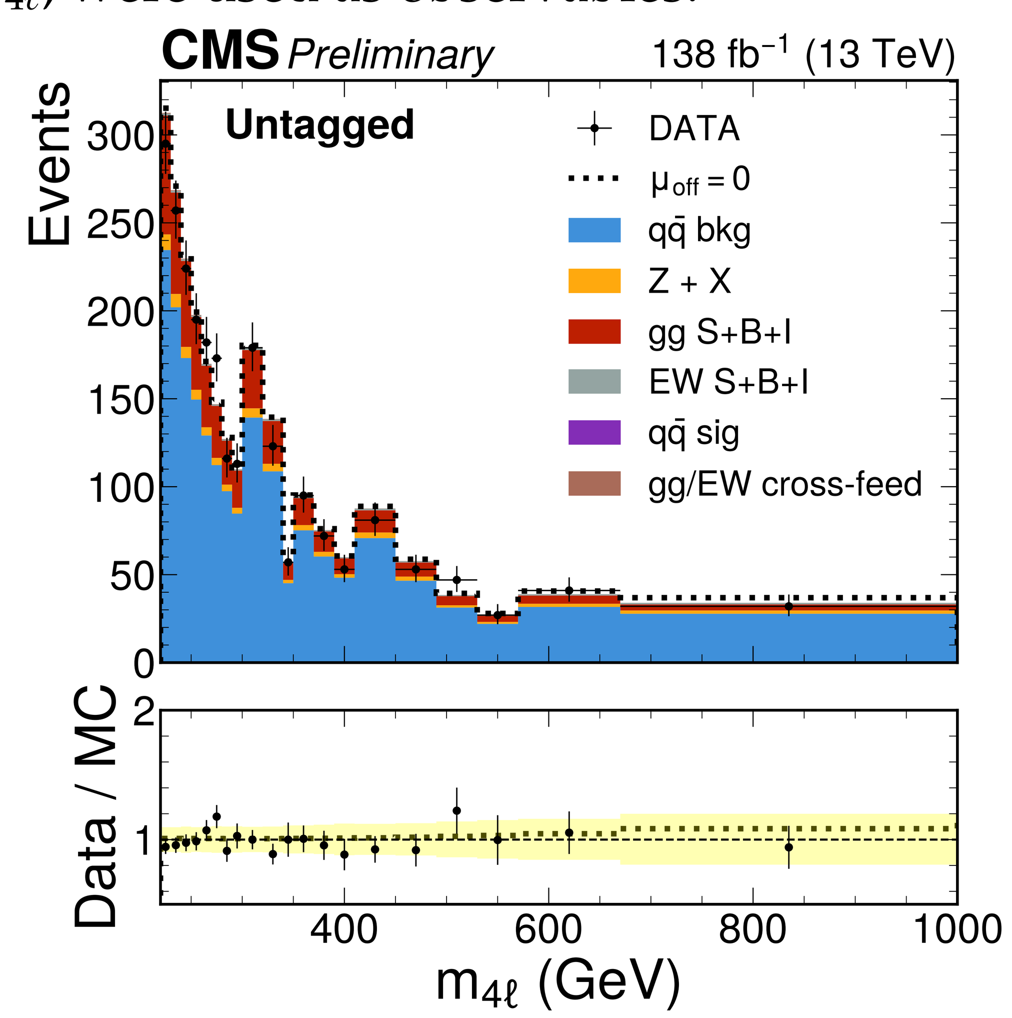
The off-shell region of the Higgs boson is enhanced in VV decays above $2m_V$, and is characterized by a strong interference component. The hallmark of the off-shell region is the independence from the the width,

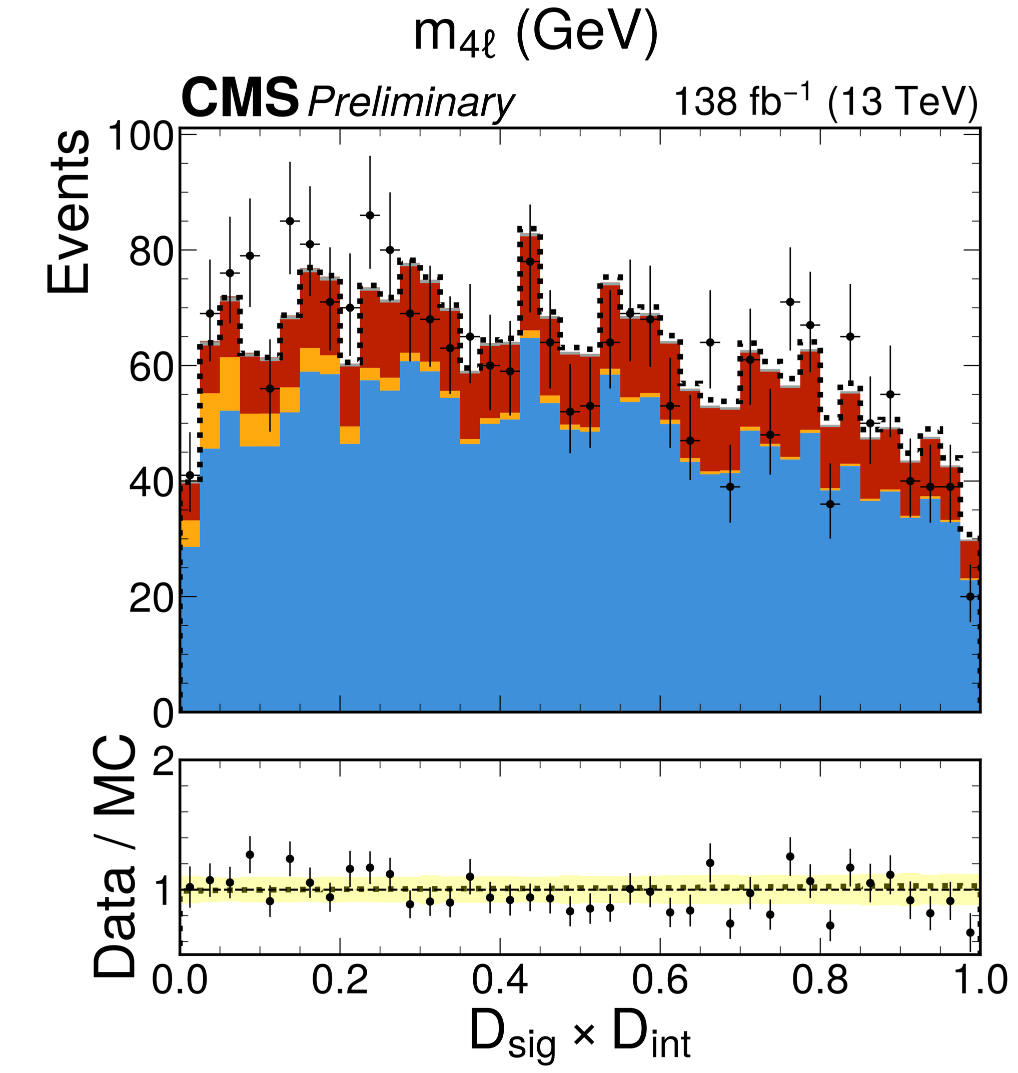
$$\frac{\sigma(i \to H \to f)}{ds} \propto \frac{1}{(s - M_H^2)^2 + M_H^2\Gamma_H^2} \quad (1)$$

$$\left(\sqrt{s} - M_H\right) >> \Gamma_H \text{ when } \sqrt{s} \geq 2m_V$$

Observables

Matrix-element discriminants, together with $m_{4\ell}$, were used as observables.

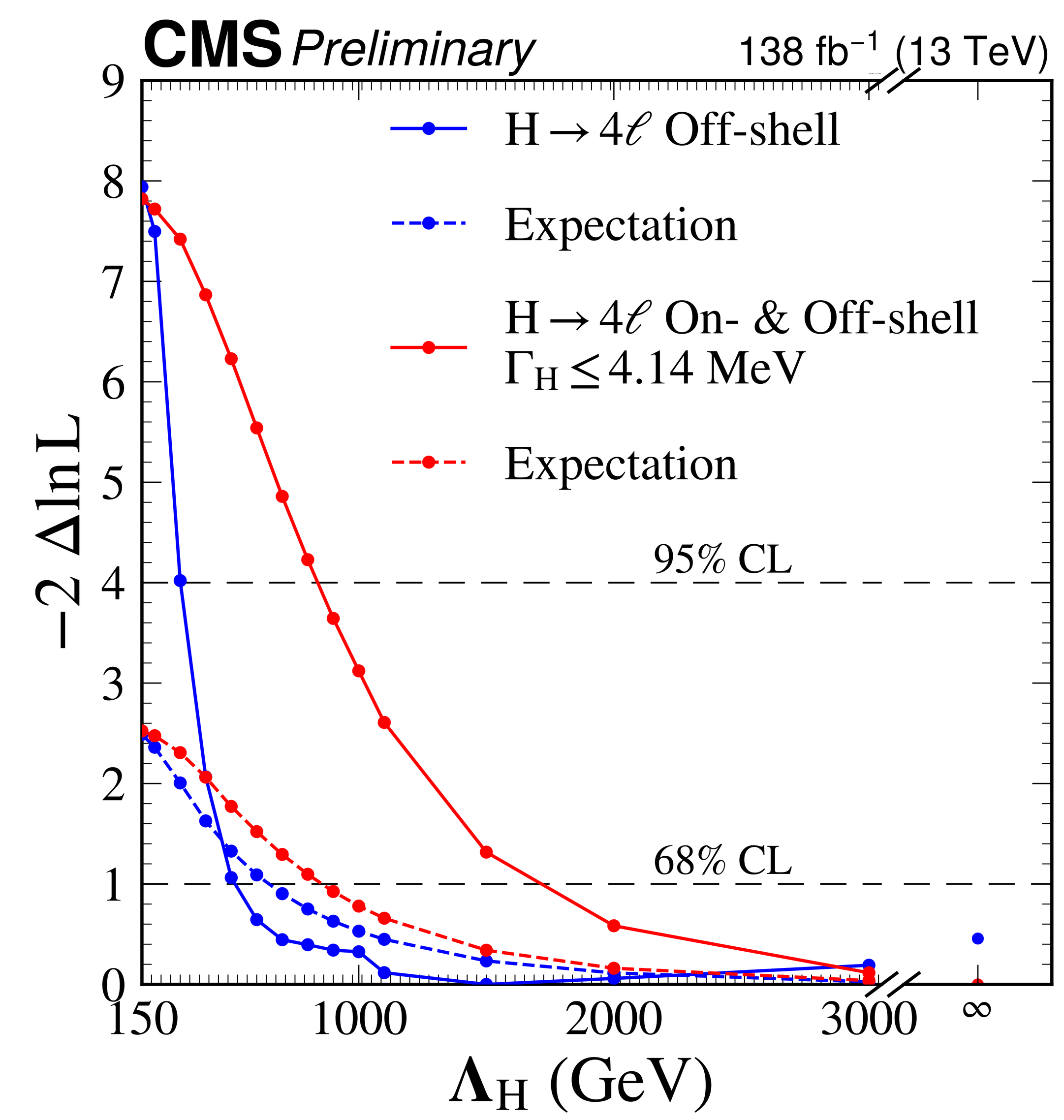




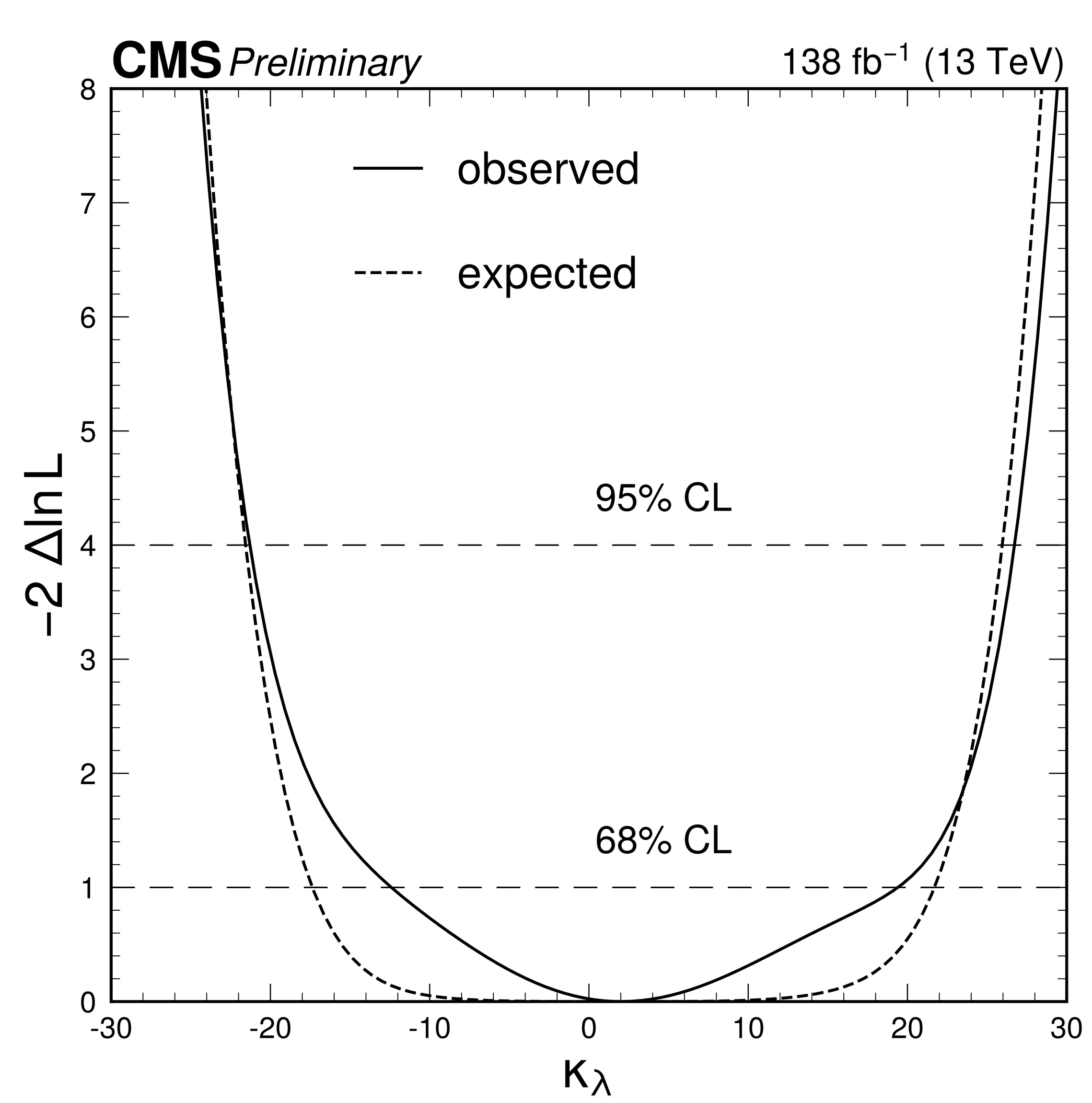
Higgs Substructure

We probe a nucleon form factor through the threshold energy Λ_H that tests Higgs boson substructure.

$$F(q^2) = \left[\frac{1}{1 + |q^2|/\Lambda_H^2}\right]^2, \qquad d = \frac{\hbar c}{\Lambda_H}$$

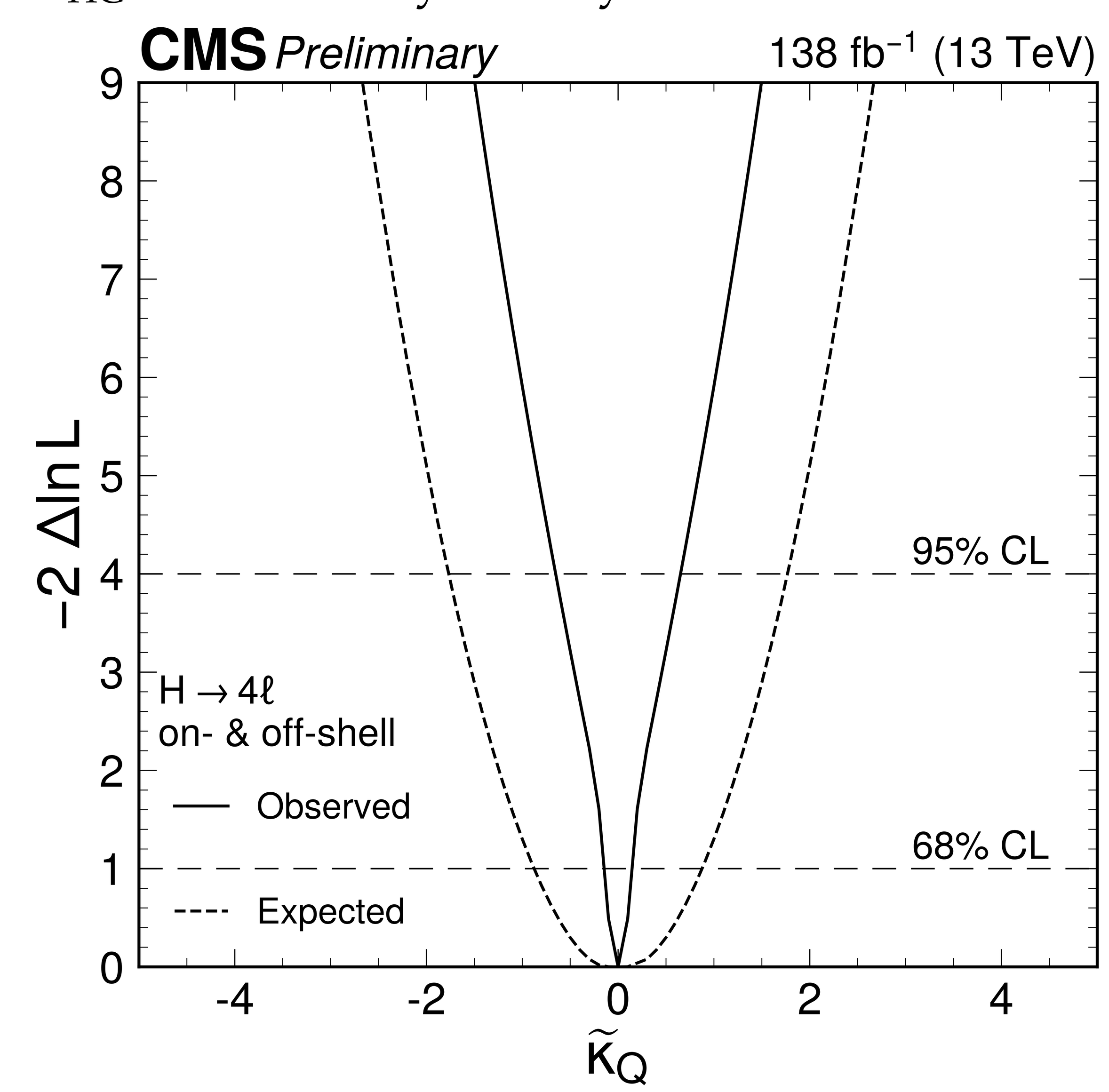


Higgs Self Coupling



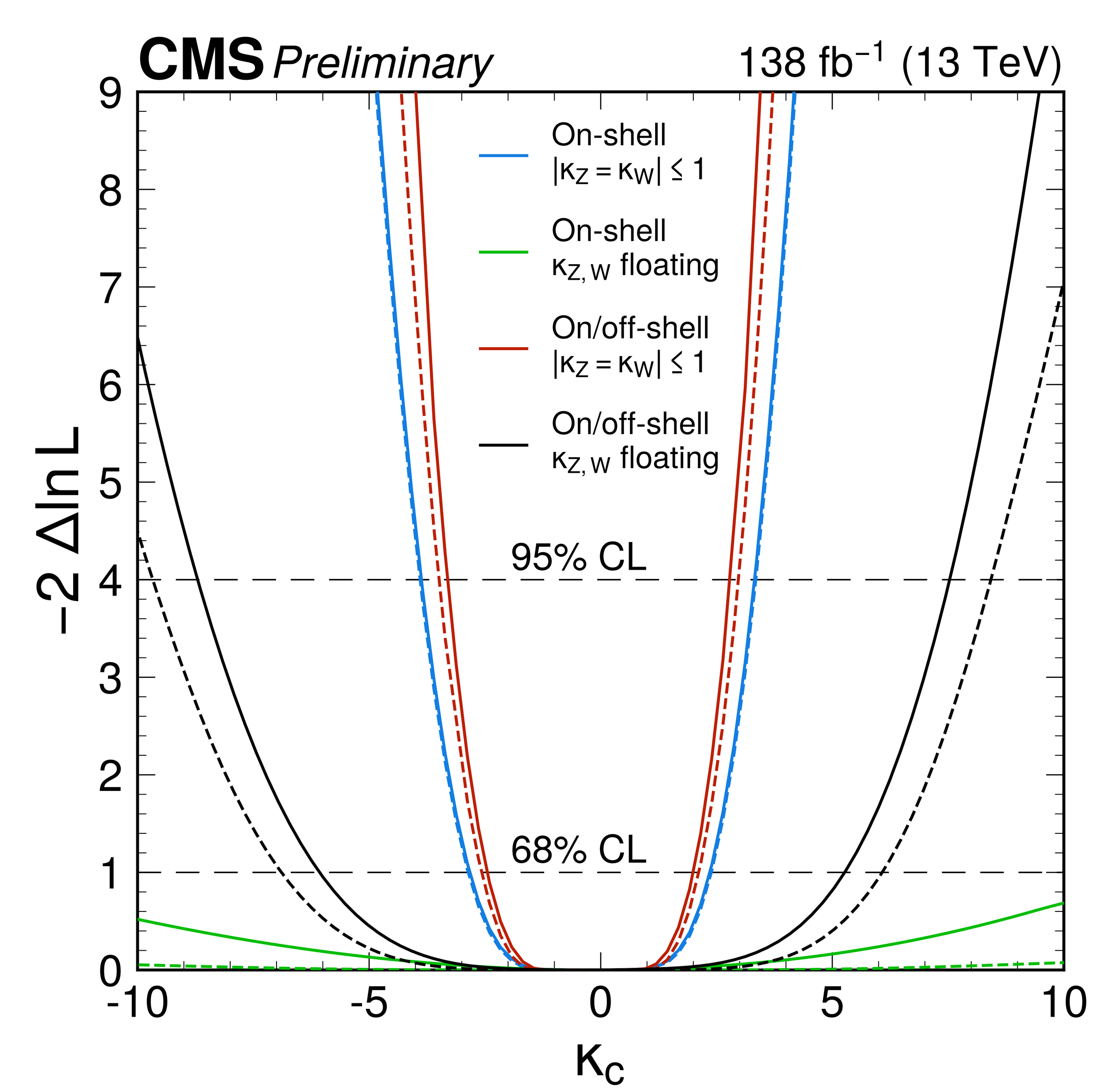
Probes of Heavy Particles

Scans over $\kappa_{t,Q}, \tilde{\kappa}_{t,Q}$ in the EFT limit within the gluon-fusion loop, where $\kappa_Q \propto C_{HG}$ and $\tilde{\kappa}_Q \propto \tilde{C}_{HG}$. Custodial symmetry was also broken.



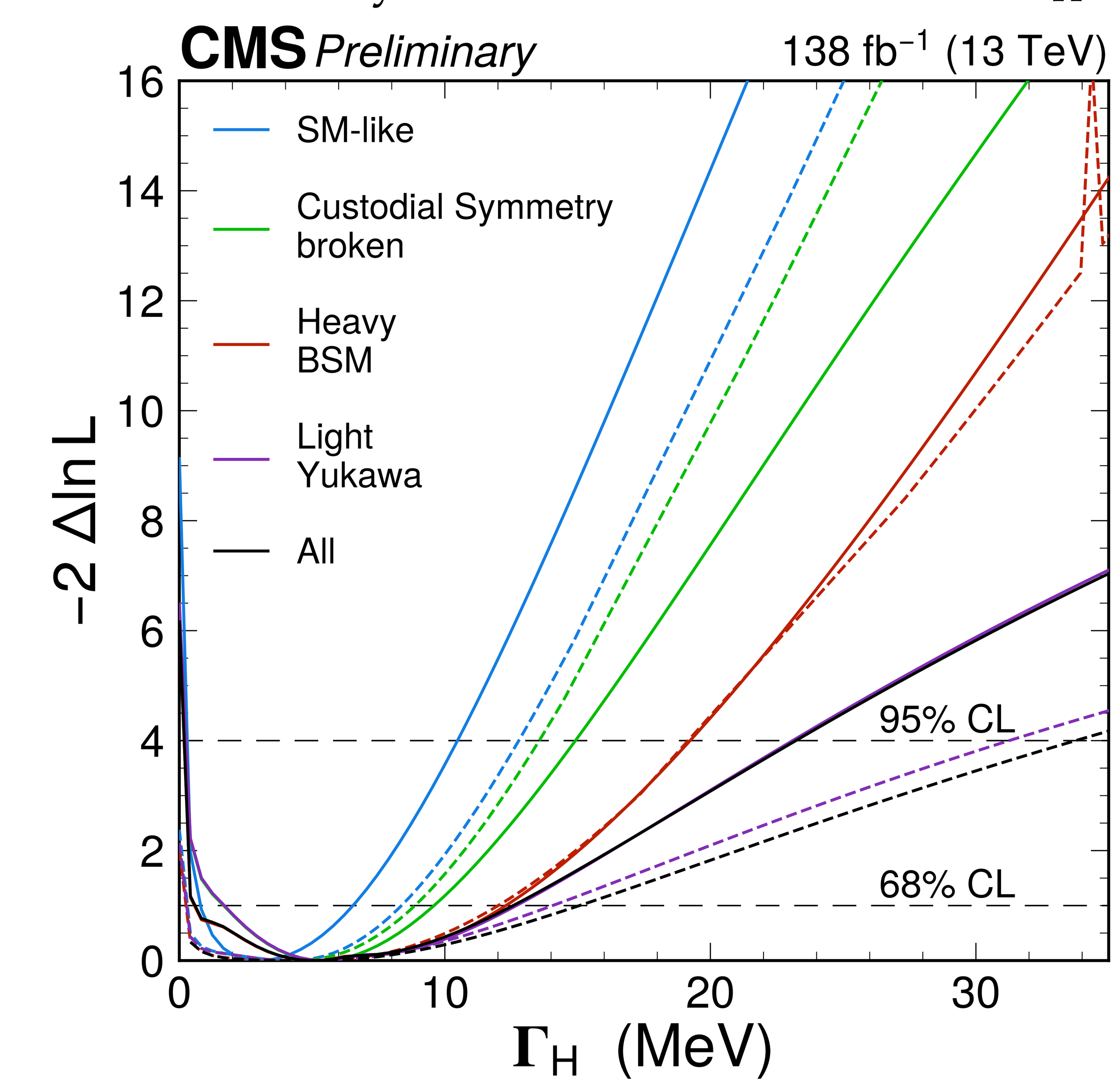
Light-quark Yukawa Couplings

Scans over u,d,s, and c quarks build off of previous work and utilize off-shell to remove constraints on vector boson couplings.



Anomalous Higgs width

Anomalous terms are allowed to float simultaneously to measure their effects on Γ_H.



Run-2 Off-shell SM Combination

A combination of vector-boson decay channels is performed for all the Run-2 off-shell analyses conducted to retrieve the most precise measurement of the SM width of the Higgs boson to date.